

SPECIAL TESTS — KNEE JOINT

I. TESTS FOR LIGAMENT INTEGRITY

A. COLLATERAL LIGAMENTS

1. VALGUS STRESS TEST

PURPOSE

To assess the integrity of the medial collateral ligament (MCL).

PROCEDURE

- Patient lies supine.
- Knee is slightly flexed, approximately 20–30°, to unlock the joint.
- Stabilize the lateral aspect of the knee.
- Apply a valgus force to the lower leg while stabilizing the thigh.

POSITIVE SIGN

- Excessive medial joint opening compared with the opposite side.
- Pain along the medial knee.

INDICATES

MCL injury/insufficiency.

2. VARUS STRESS TEST

PURPOSE

To assess the integrity of the lateral collateral ligament (LCL).

PROCEDURE

- Patient lies supine.
- Knee is slightly flexed, approximately 20–30°.
- Stabilize the medial aspect of the knee.
- Apply a varus force to the lower leg.

POSITIVE SIGN

- Excessive lateral joint opening.
- Pain along the lateral knee.

INDICATES

LCL injury/insufficiency.

B. CRUCIATE LIGAMENTS

3. LACHMAN TEST ★★★★★

PURPOSE

To assess the integrity of the anterior cruciate ligament (ACL).

PROCEDURE

- Patient lies supine.
- Knee flexed approximately 20–30°.
- Stabilize the femur with one hand.
- Hold the proximal tibia with the other hand.
- Pull the tibia anteriorly.

POSITIVE SIGN

- Excessive anterior translation of tibia.
- Soft/absent firm end-feel compared with the normal side.

INDICATES

ACL injury/tear.

Very important: Lachman is generally more sensitive for ACL injury than the anterior drawer test, particularly in the acute knee.

4. ANTERIOR DRAWER TEST

PURPOSE

To assess the ACL.

PROCEDURE

- Patient lies supine.
- Hip flexed and knee flexed to 90°.
- Stabilize the foot.
- Grasp the proximal tibia with both hands.
- Pull the tibia anteriorly.

POSITIVE SIGN

- Excessive anterior movement of the tibia compared with the opposite side.

INDICATES

ACL injury/insufficiency.

5. POSTERIOR DRAWER TEST

PURPOSE

To assess the integrity of the posterior cruciate ligament (PCL).

PROCEDURE

- Patient lies supine.
- Knee flexed to approximately 90°.
- Stabilize the foot.
- Place both hands around the proximal tibia.
- Push the tibia posteriorly.

POSITIVE SIGN

- Excessive posterior translation of the tibia.

INDICATES

PCL injury/insufficiency.

II. TESTS FOR MENISCAL INJURY

6. MCMURRAY'S TEST ★★★★★

PURPOSE

To assess for medial and lateral meniscal lesions.

PROCEDURE

- Patient lies supine.
- Fully flex the knee.
- For medial meniscus: externally rotate the tibia and apply valgus stress while extending the knee.
- For lateral meniscus: internally rotate the tibia and apply varus stress while extending the knee.
- Palpate the joint line.

POSITIVE SIGN

- Pain
- Palpable/audible click or clunk at the joint line.

INDICATES

Possible meniscal tear.

7. APLEY'S COMPRESSION & DISTRACTION TEST

PURPOSE

To differentiate meniscal pathology from ligament pathology.

PROCEDURE

Compression:

- Patient lies prone.
- Knee flexed to 90°.
- Stabilize the thigh.
- Apply downward compression through the heel.
- Rotate the tibia internally and externally.

Distraction:

- Maintain the same position.
- Apply traction to the tibia while rotating it.

POSITIVE SIGN

Compression positive:

- Pain reproduced with compression/rotation.
- → Suggests meniscal pathology.

Distraction positive:

- Pain reproduced with traction/rotation.
- → Suggests ligament/capsular pathology.

III. JOINT EFFUSION & PATELLOFEMORAL DYSFUNCTION

8. FLUCTUATION TEST

PURPOSE

To detect knee joint effusion, particularly larger effusions.

PROCEDURE

- Patient lies supine with the knee relaxed/extended.
- Examiner compresses the fluid from one side of the knee toward the opposite side.
- Observe/palpate for a fluid wave.

POSITIVE SIGN

- A palpable/visible fluid wave or fluctuation.

INDICATES

Knee joint effusion.

9. PATELLAR TAP TEST ★★★★★

PURPOSE

To detect knee joint effusion.

PROCEDURE

- Patient lies supine with knee extended.
- Gently press/compress the suprapatellar region to move fluid beneath the patella.
- Tap the patella downward against the femur.

POSITIVE SIGN

- Patella rebounds/floats and taps the femur, producing a palpable tap.

INDICATES

Significant knee joint effusion.

10. CLARKE'S TEST / PATELLAR GRIND TEST

PURPOSE

To assess patellofemoral dysfunction, particularly chondromalacia/patellofemoral pain.

PROCEDURE

- Patient lies supine with the knee extended.
- Examiner places the hand proximal to the patella.
- Apply gentle downward pressure to the patella.
- Ask the patient to contract the quadriceps.

POSITIVE SIGN

- Pain behind/around the patella.
- Inability to maintain quadriceps contraction because of pain.

INDICATES

Possible patellofemoral dysfunction/chondral pathology.

IV. OTHER MEASUREMENT

11. Q-ANGLE

PURPOSE

To assess patellofemoral alignment and the lateral pull on the patella.

MEASUREMENT

Q-angle is formed by the intersection of:

- Line 1: ASIS → centre of patella
- Line 2: centre of patella → tibial tubercle

PROCEDURE

- Patient positioned with the knee in the appropriate standardized position.
- Identify and mark:
 - ASIS
 - Centre of patella
 - Tibial tubercle
- Use a goniometer to measure the angle between the two lines.

POSITIVE/ABNORMAL FINDING

An increased Q-angle may indicate increased lateral patellar alignment/pull and may be associated with patellofemoral dysfunction.

INDICATES

- Patellofemoral malalignment
- Predisposition to lateral patellar tracking/instability

COMPLETE KNEE REVISION TABLE

Category	Test	Indicates
Collateral ligament	Valgus Stress	MCL injury
Collateral ligament	Varus Stress	LCL injury
Cruciate ligament	Lachman	ACL injury
Cruciate ligament	Anterior Drawer	ACL injury
Cruciate ligament	Posterior Drawer	PCL injury
Meniscus	McMurray's	Meniscal tear
Meniscus	Apley's Compression	Meniscal pathology
Meniscus	Apley's Distraction	Ligament/capsular pathology
Effusion	Fluctuation Test	Knee effusion
Effusion	Patellar Tap	Significant knee effusion
Patellofemoral	Clarke's Test	Patellofemoral dysfunction
Measurement	Q-angle	Patellofemoral alignment